

```
1 package Lab_07;
2
3 import java.io.File;
4 import java.util.ArrayList;
5 import java.util.Scanner;
6
7 public class DanhSachHocSinh {
8     private ArrayList<HocSinh> a = new ArrayList<>();
9
10    public void docFile(String tenFile) {
11        try {
12            File f = new File(tenFile);
13            if(f.exists()){
14                System.out.println("Co file");
15                Scanner read = new Scanner(f);
16                while(read.hasNext()){
17                    String line = read.nextLine();
18                    String s[] = line.split(",");
19                    for(int i=0;i<s.length;i++) s[i] = s[i].trim();
20                    boolean gt = Boolean.parseBoolean(s[3]);
21                    double mon1 = Double.parseDouble(s[5]);
22                    double mon2 = Double.parseDouble(s[6]);
23                    double monRieng = Double.parseDouble(s[7]);
24                    if(s[0].equalsIgnoreCase("C")){
25                        a.add(new KhoiC(s[0],s[1],s[2],gt,s[4],mon1,mon2,monRieng));
26                    }else a.add(new KhoiD(s[0],s[1],s[2],gt,s[4],mon1,mon2,monRieng));
27                }
28            }else System.out.println("Khong co file");
29        } catch (Exception e) {
30        }
31    }
32
33    public ArrayList<HocSinh> tachKhoi(String tenKhoi){
34        ArrayList<HocSinh> b = new ArrayList<>();
35        for(HocSinh i:a){
36            if(i.getClass().getSimpleName().equalsIgnoreCase(tenKhoi)) b.add(i);
37        }
38        return b;
39    }
40}
```

```
40
41 public void lietKe(ArrayList<HocSinh> b , String tieude){
42     System.out.println(tieude);
43     for(HocSinh i:b){
44         System.out.println(i);
45     }
46 }
47
48 public double DTB_khoiC(){
49     double tong=0,dem=0;
50     for(HocSinh i:a){
51         if(i instanceof KhoiC){
52             tong+=i.getDiemTrungBinh();
53             dem++;
54         }
55     }
56     return dem==0?0:tong/dem;
57 }
58
59 public double DTB_khoiD(){
60     double tong=0,dem=0;
61     for(HocSinh i:a){
62         if(i instanceof KhoiD){
63             tong+=i.getDiemTrungBinh();
64             dem++;
65         }
66     }
67     return dem==0?0:tong/dem;
68 }
69 }
```

